

David Teklea

davidgteklea@gmail.com | 702-640-6630 | LinkedIn | GitHub

EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science (ML Specialization), GPA: 4.0/4.0

Relevant Coursework: Graduate Algorithms, Deep Learning, Computer Networking, Operating Systems, Machine Learning

Atlanta, GA

Aug 2025 - Dec 2026

University of Pittsburgh

Bachelor of Science in Computer Science, GPA: 3.74/4.0

Pittsburgh, PA

Aug 2021 - May 2025

EXPERIENCE

Software Engineering Intern

Capital One

June 2026 - Aug 2026

Richmond, VA

- Owned end-to-end design and development of a nightly Python and AWS Lambda reconciliation pipeline that enriched governance data in an internal PostgreSQL catalog of thousands of applications by syncing 481 ITSM records from Snowflake, eliminating eight months of data staleness.
- Reduced Lambda execution time 61%, from over 300 seconds to 116 seconds, by diagnosing VPC DNS and proxy conflicts and moving connection pooling and AWS Secrets Manager retrieval outside record-processing loops, achieving 100% success for found assets.
- Extended Flask REST APIs and Flyway-managed PostgreSQL schemas with 89.4% unit test coverage to surface ownership and key-asset governance data in CSV and BI dashboards, enabling enterprise stakeholders to identify responsible teams.
- Productionized the pipeline across development, QA, and production using Jenkins, CloudFormation, IAM roles, and Secrets Manager, then migrated its Snowflake authentication to OAuth 2.0 before an enterprise security cutoff with zero downtime.

Machine Learning Researcher, Dr. Xiang Lorraine Li Group

University of Pittsburgh

June 2025 - Dec 2025

Pittsburgh, PA

- Built a configuration-driven Python and OpenAI API pipeline that parsed four verb spreadsheets and generated and rated 885 scenarios across 177 verbs for semantic-role experiments, dynamically including Agent, Patient, Instrument, and Location roles.
- Integrated IBM's Transition AMR Parser to convert scenarios into Penman graphs and extract semantic roles to drive performance up 2%.
- Built an evaluation harness on 100 human-annotated scenarios that measured precision, recall, and macro-F1, providing quantitative insight into parser performance.
- Cut end-to-end runtime over 50% and saved about \$100 by splitting LLM generation from AMR parsing and parallelizing Slurm batch jobs on Pitt's HPC cluster.

Software Engineering Intern

PNC

June 2024 - Aug 2024

Pittsburgh, PA

- Increased FastAPI service test coverage over 30% by refactoring Pydantic models and adding pytest regression tests to catch API schema regressions before deployment.
- Built Python and Automation Anywhere bots to automate monthly AML case-assignment distribution and shared-drive cleanup, supporting investigators handling hundreds of cases and managing hundreds of files.

STEP Intern

Google

May 2023 - Aug 2023

Cambridge, MA

- Built Java and TypeScript URL, country, and language prefill flows for Google's content-removal product, routing requests across 10+ legal and non-legal policy forms spanning billions of indexed results, which streamlined the removal process.
- Implemented experiment feature flags with Mendel for controlled rollout and added unit and compatibility tests across desktop and mobile, achieving 70% test coverage.
- Built a responsive Soy and SCSS accordion to replace a request-details dialog and validated eight expanded and collapsed states with SCUBA visual-regression tests, resulting in a more consistent user interface.

PROJECTS

EigenDB

C, C++, Go, API Gateway, S3

- Built a C and C++ vector search engine with AVX2 SIMD distance kernels and HNSW approximate nearest-neighbor indexing across 100,000+ vectors.
- Implemented S3 snapshot persistence in a Go API using AWS SDK v2, restoring collections of 500K+ vectors in under two seconds and enforcing rate limits through API Gateway.

SKILLS

Languages: Python, Java, TypeScript, C, C++, Go, SQL, Bash, HTML/CSS, SCSS, Soy

Backend & Cloud: Flask, FastAPI, REST APIs, Lambda, API Gateway, S3, CloudWatch, IAM, Secrets Manager, CloudFormation, OAuth 2.0

Data & ML: PostgreSQL, Snowflake, Flyway, Pandas, PyTorch, scikit-learn, Pydantic, Slurm

Testing & Tools: pytest, unittest, Jenkins, Docker, Git, Linux